Junhyung Lyle Kim

Legal name: Junhyung Kim

jlylekim.github.io | google scholar | jlylekim | jlylekim | jlylekim

Employment

JPMorganChase, Global Technology Applied Research — New York, NY
Quantum Computing Research Scientist — Sr. Associate — Oct 2024 - Present

- Research interests: optimization; quantum algorithms; randomized algorithms; machine learning

Education

Rice University — Houston, TX
Ph.D. in Computer Science — Aug 2019 - Aug 2024

- Advisor: Prof. Anastasios Kyrillidis `[website]`; Committee members: Profs. César A. Uribe and Nai-Hui Chia
- Thesis: Algorithmic and Structural Acceleration Techniques in Machine Learning and Quantum Computing

University of Chicago — Chicago, IL
B.A. in Mathematics; B.A. in Statistics — Oct 2013 - Jun 2017

- Advisor: Prof. Panos Toulis `[website]`; General Honors; Dean's List 2013 - 2017

Professional Experience

JPMorganChase, Global Technology Applied Research — New York, NY
Quantum Computing Research Intern; Host: Dr. Marco Pistoia — Jun 2024 - Aug 2024

- Design, analysis, and application of quantum / quantum-inspired classical algorithms

Mila – Quebec Artificial Intelligence Institute — Montréal, QC
Visiting Student Researcher; Hosts: Profs. Ioannis Mitliagkas and Gauthier Gidel — May 2023 - Aug 2023

- First-order methods for variational inequality in function space / curvature adaptive optimization algorithm

Meta, Fundamental AI Research (FAIR) — New York, NY
AI Research Intern; Host: Dr. Aaron Defazio — May 2022 - Aug 2022

- Theory and application of adaptive stochastic gradient methods for deep learning

Republic of Korea Special Warfare Training Group (SWTG) — Gyeonggi, South Korea
Sergeant — Jan 2012 - Oct 2013

- Airborne training (certified paratrooper license #748-416); maritime infiltration training

Academic Experience

Rice University, Computer Science Department — Houston, TX
Ph.D. Candidate; Advisors: Profs. Anastasios Kyrillidis, César A. Uribe, and Nai-Hui Chia — Aug 2019 - Aug 2024

- Active collaborations with Google (F. Pedregosa) and IBM (G. Kollias) on optimization and quantum computing
- Internships at Meta (FAIR), MILA – Quebec AI Institue, and JPMorganChase (quantum computing)

University of Chicago, Booth School of Business — Chicago, IL
Research Assistant; Supervisors: Profs. Panos Toulis and Sanjog Misra — Jun 2017 - Jul 2019

- Stochastic approximation for large-scale inverse reinforcement learning

University of Chicago, Statistics Department — Chicago, IL
Research Assistant; Supervisors: Profs. Mikael Kuusela and Michael Stein — Oct 2016 - Jun 2017

- Uncertainty quantification for high-energy physics unfolding problem; `[code]`; `[documentation]`

Publications

(* denotes equal contributions)

Journal Papers

[J1] Towards Quantum Advantage in Global Optimization of Nonconvex Functions
G. Ozgul*, D. Herman*, A. Apte, **J. L. Kim**, A. Prakash, J. Shen, S. Chakrabarti
Nature Communications 2026
Also presented at QIP 2026

[J2] On Speedups for Convex Optimization via Quantum Dynamics
S. Chakrabarti*, D. Herman*, J. Watkins*, E. Fontana, B. Augustino, **J. L. Kim**, M. Pistoia
Theory of Quantum Computation, Communication and Cryptography, ***TQC 2026***
Also presented at QSim 2025

[J3] How Much Pre-training Is Enough to Discover a Good Subnetwork?
C. Wolfe*, F. Liao*, Q. Wang, **J. L. Kim**, A. Kyrillidis
Transactions on Machine Learning Research, ***TMLR 2024***

[J4] When is Momentum Extragradient Optimal? A Polynomial-Based Analysis
J. L. Kim, G. Gidel, A. Kyrillidis, F. Pedregosa
Transactions on Machine Learning Research, ***TMLR 2024***
Workshop on Optimization for Machine Learning, NeurIPS 2022

[J5] Fast Quantum State Reconstruction via Accelerated Non-Convex Programming
J. L. Kim, G. Kollias, A. Kalev, K.X. Wei, A. Kyrillidis
Photonics 2023

[J6] Local Stochastic Factored Gradient Descent for Distributed Quantum State Tomography
J. L. Kim, M. T. Toghani, C. A. Uribe, A. Kyrillidis
Control Systems Letters, ***L-CSS 2022***

Conference Papers (peer-reviewed)

[C1] Learning Rate Decay Can Exponentially Accelerate SGD for Global Optimization of Nonconvex Functions
G. Ozgul*, D. Herman*, **J. L. Kim**, S. H. Sureshbabu, S. Chakrabarti
Advances in Neural Information Processing Systems, ***NeurIPS 2026***

[C2] A Catalyst Framework for the Quantum Linear System Problem via the Proximal Point Algorithm
J. L. Kim, N. H. Chia, A. Kyrillidis
AAAI Conference on Artificial Intelligence, ***AAAI 2026 (Oral)***

[C3] Fast Zeroth-Order Convex Optimization with Quantum Gradient Methods
J. L. Kim*, B. Augustino*, D. Herman*, E. Fontana*, J. Watkins, M. Pistoia, S. Chakrabarti
Advances in Neural Information Processing Systems, ***NeurIPS 2025***

[C4] Solving Hidden Monotone Variational Inequalities with Surrogate Losses
R. D'Orazio, D. Vucetic, Z. Liu, **J. L. Kim**, I. Mitliagkas, G. Gidel
International Conference on Learning Representations, ***ICLR 2025***
Workshop on Optimization for Machine Learning, NeurIPS 2024

[C5] On the Error-Propagation of Inexact Hotelling's Deflation for Principal Component Analysis
F. Liao, **J. L. Kim**, C. Barnum, A. Kyrillidis
International Conference on Machine Learning, ***ICML 2024***

[C6] Adaptive Federated Learning with Auto-Tuned Clients
J. L. Kim, M. T. Toghani, C. A. Uribe, A. Kyrillidis
International Conference on Learning Representations, ***ICLR 2024***
Workshop on Federated Learning and Analytics in Practice, ICML 2023

[C7] Convergence and Stability of the Stochastic Proximal Point Algorithm with Momentum
J. L. Kim, P. Toulis, A. Kyrillidis
Conference on Learning for Dynamics and Control, ***L4DC 2022***
Workshop on Optimization for Machine Learning, NeurIPS 2021 (Spotlight)

Preprints

(* denotes equal contributions)

[P1] The Quartic Gap Between Quantum and Dequantized Algorithms for Linear Algebra Is Tight
S. Chakrabarti, D. Herman, **J. L. Kim**, G. Ozgul

[P2] Hypergraph Cycle Enumeration for Efficient Inference of Planted k-XOR with Smooth Trade-off
A. Babakhani, E. Fontana, **J. L. Kim**, S. Chakrabarti, R. Shaydulin

[P3] From Quantum Tunneling to Classical Optimization through Instantons
J. Shen, D. Herman, G. Ozgul, A. Seshadri, **J. L. Kim**, S. H. Sureshbabu, S. Chakrabarti, N. Kumar

[P4] Quantum Simulation on Riemannian Manifolds
D. Herman, J. Watkins, G. Ozgul, J. Shen, B. Augustino, **J. L. Kim**, S. Chakrabarti

[P5] Near-Optimal Quantum Lower Bounds for Convex Optimization via Fourier Rank
B. Augustino, S. Chakrabarti, E. Fontana, D. Herman, **J. L. Kim**, G. Ozgul, N. Voronova

[P6] Faster Algorithms for Multimarginal Optimal Transport
B. Augustino, Y. Sun, A. Acharya, S. Chakrabarti, **J. L. Kim**, S. H. Sureshbabu, C. Che

[P7] Anytime Training with Schedule-Free Spectral Optimization
A. Apte, P. Deshpande, N. Kumar, S. Chakrabarti, **J. L. Kim**

[P8] Fault-Tolerant End-to-end Quantum Algorithms for Tensors
E. Fontana, S. Prasad, **J. L. Kim**, J. Sullivan, M. Perlin, R. Shadlyn, S. Chakrabarti
TQC 2026

[P9] A Simple Analysis of a Quantum-Inspired Algorithm for Solving Low-Rank Linear Systems
T. Chen*, **J. L. Kim***, A. Ray*, S. Chakrabarti, D. Herman, N. Kumar

[P10] Smoothness-Adaptive Sharpness-Aware Minimization for Finding Flatter Minima
H. Naganuma*, **J. L. Kim***, A. Kyrillidis, I. Mitliagkas
Workshop on Practical Machine Learning for Low Resource Settings, ICLR 2024

Honors & Awards

2026 ICML 2026 Silver Reviewer
2025 NeurIPS 2025 Top Reviewer (complimentary registration)
2025 ICLR 2025 Notable Reviewer
2024 Rice Engineering Alumni Graduate Student Spring Travel Grant ($540)
2023 Rice Engineering Alumni Graduate Student Fall Travel Grant ($480)
2023 AISTATS 2023 Top Reviewer (Top 10 %, complimentary registration)
2022 Rice Engineering Alumni Graduate Student Fall Travel Grant ($1,200)
2022 Rice Engineering Alumni Graduate Student Spring Travel Grant ($960)
2021 Spotlight paper, Workshop on Optimization for Machine Learning (NeurIPS 2021)
2021 Rice Engineering Alumni Graduate Student Fall Travel Grant ($1,900)

Service

Workshops QuantIPS 2023: Co-organizer for "Quantum Information Processing Systems" [link]
TL;DR 2023: Co-organizer for "Texas Colloquium on Distributed Learning" [link]
ICML 2021: Co-organizer for "Beyond First Order Methods in Machine Learning Systems" [link]
Reviews Quantum, QEC, TMLR, NeurIPS, ICML, ICLR, AISTATS

Mentorship

Undergraduate students

Co-advised with Prof. Anastasios Kyrillidis

- Rithik Jain (Rice University): sparse learning with hadamard product — Mar 2021 - May 2022
- Justin Lumpkin (U of Maryland): deep matrix factorization; Google/Rice REU 1st place — May 2021 - Aug 2021
- Cruz Barnum (Reed College): scalable streaming PCA; Google/Rice REU 2nd place — May 2021 - Aug 2021

Others

Leadership *President*, Rice University Computer Science Graduate Student Association (2022 - 2023)
President, UChicago Korean Undergraduate Maroon Association (2016 - 2017)
Software MiFGD (Python) [link], sgd (R package) [link], UndersmoothedUnfolding (C++) [link]
Language Korean (native), English (bilingual proficiency)

Invited Talks

Adaptive Federated Learning with Auto-Tuned Clients — Phoenix, AZ
Annual Meeting, INFORMS — Oct 2023

Adaptive Federated Learning with Auto-Tuned Clients — Montréal, Canada
Montréal Machine Learning and Optimization (MTL MLOpt), MILA — Jun 2023

Local Stochastic Factored Gradient Descent for Distributed Quantum State Tomography — Cancún, Mexico
IEEE Conference on Decision and Control (CDC) — Dec 2022

Convergence and Stability of the Stochastic Proximal Point Algorithm with Momentum — Indianapolis, IN
Optimization for Machine Learning, INFORMS — Oct 2022

Convergence and Stability of the Stochastic Proximal Point Algorithm with Momentum — Bethlehem, PA
International Conference on Continuous Optimization (ICCOPT) — Jul 2022

Fast Quantum State Reconstruction via Accelerated Non-Convex Programming — Houston, TX
Quantum Group Meeting Seminar, Rice University — Jan 2022

Acceleration and Stability of the Stochastic Proximal Point Algorithm — Virtual
Workshop on Optimization for Machine Learning, NeurIPS — Dec 2021

Fast Quantum State Reconstruction via Accelerated Non-Convex Programming — Anaheim, CA
Optimization in Quantum Computing, INFORMS — Oct 2021

Other Experience

Dimensional Fund Advisors — Austin, TX

Research Intern, Investment Analytics & Data Group — Jun 2016 - Sep 2016

- Automated checking system development for security database; VBA tool for data comparison and visualization

Cook M&A Advisory Services — Chicago, IL

Investment Banking Summer Analyst — Jun 2015 - Aug 2015

- Data analysis for several buy-side projects; client document drafting

Freenters, Inc. Chicago, IL
Operations Intern Aug 2014 - Jan 2015

- VBA tool for automatically personalized email dispatching; logo/poster design (Adobe Illustrator)